LE-HUYNH TRUC-LY

I have over five years of experience in data analysis across projects spanning environmental science, health science, political science, and biology. My passion lies in data analysis, visualization, automation, and creating digital tools for research. This very CV was even crafted using code.

PROJECTS

2024 | Present

Climate and Health in Virginia: Investigating Underlying Causes of Climate-Related Diseases and Disparities

Environmental Institute, University of Virginia — Virginia, USA

- Skills used — Big data analytics, Meta-analysis, Geospatial health modelling, R, RMarkdown, Git, GNU Make
- Analyzing large-scale patient-level health records, environmental, and demographic datasets to identify unequal health risks during climate extremes, particularly extreme heat events

2025 | Present

Does Urban Green Space Mitigate Emergency Department Cases During Heat Waves?

Environmental Institute, University of Virginia — Virginia, USA

- Skills used — Big data analytics, Meta-analysis, Geospatial health modelling, R, RMarkdown, Git, GNU Make
- Analyzing large-scale patient-level health records, ERA5-Land reanalysis, and landscape metrics to quantify protective effects of neighborhood green spaces during extreme heat events

2024 | Present

Local-global linkages in biodiversity governance

Wyss Academy for Nature, University of Bern — Bern, Switzerland

- Skills used — Network analysis, Text Mining analysis, Discourse analysis, R, RMarkdown, Git, GNU Make
- Analyzing large-scale governance network data to assess whether the global biodiversity regime delivers transformative change

2023 | Present

Toward personalized acute myeloid leukemia (AML)

Biotech Research & Innovation Centre (BRIC), University of Copenhagen — Copenhagen, Denmark

- Skills used — Bioinformatics, R, RMarkdown, Git, GNU Make
- Analysed Mass Spectrometry-based Proteomics data to identify novel druggable oncogenic signaling molecules in a genetically engineered AML mouse model
- Analyzing Mass Spectrometry-based Proteomics data from AML patients to identify a broad spectrum of relevant druggable key signalling molecules that can be targeted in a clinical setting for guided personalized AML treatment

CONTACT

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SKILLS

Data Science: Frequentist and Bayesian Inference, Bayesian Modelling, Biostatistics, Bioinformatics (Proteomics data analysis)

Programming: R (advanced: published 3 R-packages), Python, SAS, JAGS, Stan

Reproducible Report: Markdown/RMarkdown, Quarto, LaTeX

DevOps: Git, GNU Make, Docker

Visualization: ggplot2, plotly, leaflet, Shiny

Laboratory: Molecular Biology, Molecules Analysis, Water Quality Analysis

Languages: Vietnamese (native), English (fluent), Japanese (elementary)

2021 | 2022

Development of predictive biomarker spectrum for neuropsychiatric systemic lupus erythematosus (NPSLE)

Graduate School of Biomedical Sciences, Nagasaki University — Nagasaki, Japan

- Skills used — Bayesian Modelling, R, SAS, RMarkdown, LaTeX, Git, GNU Make
- Estimated the cutoff concentration of anti-suprabasin antibodies in a huge patient database to predict NPSLE, through utilizing a novel Bayesian model
- Implemented reproducible workflow for data analysis and data visualization

2020 | 2023

Bayesian predictive model for toxic cyanobacteria occurrence from eutrophication and climate data

Graduate School of Engineering, Nagasaki University — Nagasaki, Japan

- Skills used — Bayesian Modelling, R, JAGS, Stan, RMarkdown, LaTeX, Git, GNU Make
- Developed Bayesian model addressed zero-inflation issue in cyanobacterial data, predicting presence probability, abundance, and WHO alert level exceedance for toxic cyanobacteria
- Improved overall modelling efficiency through the implementation of Parallel Processing techniques
- Automated the data analysis and reporting system using GNU Make, R, and Rmarkdown

EDUCATION

2018 | 2023

Doctor of Philosophy - PhD in Water and Environmental Science

Nagasaki University — Nagasaki, Japan

- Dissertation: Statistical Investigation into the Effects of Climate and Eutrophication on the Occurrence of Cyanobacteria in Small Ponds and Reservoirs

2016 | 2018

Master of Engineering - MEng in Water and Environmental Engineering

Nagasaki University — Nagasaki, Japan

- Thesis: Statistical Analysis on the Relationship among Environmental Factors, Microcystin Synthesis Gene, and Microcystin Degradation Gene

2012 | 2016

Bachelor of Science - BS in Environmental Engineering Technology

Vietnam National University – Ho Chi Minh City (VNU-HCM) University of Science — Ho Chi Minh City, Vietnam

- Thesis: Research on Sewage Sludge Dewatering System using Solar Energy

WORK EXPERIENCE

Postdoctoral Research Associate

Environmental Institute, University of Virginia, USA — 2024 - Present

Visiting Researcher

Graduate School of Engineering, Nagasaki University, Japan — 2024 - 2025

Independent Data Analyst

Self-employed at lehuynh.rbind.io — 2021 - Present

- Services: Reproducible data analysis (in environmental science, health science, biology, bioinformatics), data visualization, cartography, data communication, consulting, R package development
- Tools: R, SAS, Python, ggplot2, plotly, leaflet, Shiny

- **Project Researcher**
  Graduate School of Engineering, Nagasaki University, Japan — 2023
- **Research Assistant**
  Biological Treatment and Ecological Engineering Laboratory, Nagasaki University, Japan — 2018 - 2019
- **Technical Assistant**
  Biological Treatment and Ecological Engineering Laboratory, Nagasaki University, Japan — 2017 - 2019
- **Graduate Teaching Assistant**
  Graduate School of Engineering, Nagasaki University, Japan — 2018 - 2020

ADDITIONAL INFORMATION

R Package Development

- **chva.extras**: Supplementary tools for climate and health research in Virginia. **Le-Huynh, T.-L.**, Davis, R., Novicoff, W., DeGuzman, P. (2025)
- **omics4drug**: An R toolkit to facilitate Mass Spectrometry-based Proteomics and Phosphoproteomics data analysis. Nguyen, T.-K. Y., **Le-Huynh, T.-L.** (2025)
- **lehuynh**: Personal R package. **Le-Huynh, T.-L.** (2023)

Honours and awards

- Planetary Health Research Fellowship, Nagasaki University, Japan (2022 – 2023)
- Asian Student Foundation Scholarship, Asian Student Foundation, Japan (2017 – 2019)
- Monbukagakusho Honors Scholarship, Japan Student Services Organization, Japan (2016 – 2017)
- CHEER for Vietnam Scholarship Award for Innovation and Creativity, CHEER for Vietnam Organization, USA (2015)

Grants and Funding

- **Spark Grant**, *Investigator*, funded by Environmental Institute, University of Virginia — USD 30,000 for project *Does Urban Green Space Mitigate Emergency Department Cases During Heat Waves?*

Service & Community Engagement

- Translated R cheat sheets (Git & GitHub, gtsummary, RStudio IDE) into Vietnamese
- Member of National Center for Faculty Diversity and Development, USA (2025 - present)
- Member of Virginia Women in High Performance Computing, USA (2024 - present)
- Member of US National Postdoctoral Association, USA (2024 - present)
- Member of Japan Society on Water Environment, Japan (2017 - 2023)

Publications

- Nguyen, V. T. H., **Le-Huynh, T.-L.**, Gadola, S., Nguyen, Q., Walters, G., Owuor, M. A (2025). Local-Global Linkages in Biodiversity Governance: The Regime Complex of the Convention on Biological Diversity Agenda for Nature Pledges. *Environmental Science & Policy*, 173(November), 104246.
- Saenchan, S., Shimizu, K., Iwami, N., Angalika, M. W. S., Maseda, H., Minh, N. B., Vu, H. V., **Le-Huynh, T.-L.**, Itayama, T. (2025). Effects of pH and ultrasonication for decomposition of Microcystis formed colony in a bloom sample from a reservoir. *Desalination and Water Treatment*, 322, 101203.
- Huynh, V. V., Nguyen, M. B., Somsri, S., **Le-Huynh, T.-L.**, Ueyama, T., Shirayanagi, S., Itayama, T. (2023). Investigation of performance in MBR operated with low DO for low C/N ratio wastewater. *Water, Air, & Soil Pollution*. 235(540).
- **Le-Huynh, T.-L.**, Itayama, T., Mitsunaga, K., Angalika, M., Suzuki, S. (2022). Application of hurdle Poisson model to predict the abundance of toxic cyanobacteria Microcystis in reservoirs. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 47–51.
- **Le-Huynh, T.-L.**, Iwami, N., Whangchai, N., Gutierrez, R., Shimizu, K., Itayama, T. (2022). Statistical analysis of the effects of environmental factors and fish species on class-sorted phytoplankton composition in aquaculture ponds in northern Thailand. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 32–38.
- Angalika, M. W. S., Suzuki, S., **Le-Huynh, T.-L.**, Itayama, T., Tanaka, W. (2022). Assessing nutrient budget of ungauged catchment using intermittent water quality markers. *Maejo International Journal of Energy and Environmental Communication*, 4(3), 1–10.
- Hoang, T. T. T., Ichinose, K., Morimoto, S., Furukawa, K., **Le-Huynh, T.-L.**, Kawakami, A. (2022). Measurement of anti-suprabasin antibodies, multiple cytokines and chemokines as potential predictive biomarkers for neuropsychiatric systemic lupus erythematosus. *Clinical Immunology*, 237(March), 1–8.

Selected Presentations

- **Le-Huynh, T.-L.**, Novicoff, W., DeGuzman, P., Davis, R., Exploring climate-health disparities in Virginia: Toward AI-driven solutions [Poster presentation], *Environmental Futures Forum 2025*, Charlottesville, Virginia, October 2025.
- **Le-Huynh, T.-L.**, Iwami, N., Praphrute, R., Whangchai, N., Gutierrez, R., Shimizu, K., Itayama, T., Statistical analysis on phytoplankton population at hypertrophic ponds in northern Thailand [Oral presentation], *The 57th Annual Conference of Japan Society on Water Environment*, Ehime, Japan, March 2023.
- **Le-Huynh, T.-L.**, Itayama, T., Mitsunaga, K., Angalika, M., Suzuki, S., Predict toxic cyanobacteria Microcystis in reservoirs by Bayesian hurdle Poisson model [Oral presentation], *1st Campus Asia Program International Symposium*, Nagasaki, Japan, February 2023.
- **Le-Huynh, T.-L.**, Itayama, T., Mitsunaga, K., Using Bayesian hurdle Poisson model to predict cyanobacterial cell densities in Nagasaki reservoirs [Oral presentation], *The 56th Annual Conference of Japan Society on Water Environment*, Toyama, Japan, March 2022.
- **Le-Huynh, T.-L.**, Mitsunaga, K., Itayama, T., A Bayesian model for predicting the growth of toxic Microcystis from air temperature and trophic state index [Oral presentation], *The 3rd International Conference on Renewable Energy, Sustainable Environmental and Agricultural Technologies*, Chiangmai, Thailand, December 2021.
- **Le-Huynh, T.-L.**, Itayama, T., Nguyen, T. H. G., Xia, D., Shimizu, K., Iwami, N., Okano, K., Maseda, H., Praphrute, R., Ruangdet, K., Gutierrez, R., Whangchai, N., Influence of environmental factors on Microcystins degradation bacteria and toxigenic cyanobacteria bloom: a Bayesian approach [Poster presentation], *The NaToxAq Conference on Natural Toxins: Environmental Fate & Safe Water Supply*, Brno, Czech Republic, September 2020.
- **Le-Huynh, T.-L.**, Giang, N. T. H., Xia, D., Shimizu, K., Iwami, N., Okano, K., Maseda, H., Reunkaew, P., Whangchai, N., Itayama, T., Bayesian analysis on environmental factors for standing crops of microcystin, synthesis gene and degradation gene in eutrophicated ponds in Thailand [Oral presentation], *The 53rd Annual Conference of Japan Society on Water Environment*, Yamanashi, Japan, March 2019.
- **Le-Huynh, T.-L.**, Huong, N. T. G., Xia, D., Shimizu, K., Iwami, N., Okano, K., Maseda, H., Reunkaew, P., Whangchai, N., Itayama, T., Quantitative analysis on cyanobacteria related genes presence in aquaculture ponds at different elevation in Thailand [Oral presentation], *The 52nd Annual Conference of Japan Society on Water Environment*, Hokkaido, Japan, March 2018.